

Online Appendix

Chained Indices Unchained: Structural Transformation and the Welfare Foundations of Income Growth Measurement

Omar Licandro

University of Leicester

Juan Ignacio Vizcaino

University of Nottingham

September 2026

This Appendix is intended for online publication only. Appendix A contains the proofs of Propositions 1 to 5. Appendix B verifies the regularity condition of Lemma 1 in [Boppart \(2014\)](#) along the equilibrium path and for the preference orderings underlying the fixed-base indices. Appendix C applies the welfare measures to an economy without capital. Appendix D characterizes the two-stage structure of the equivalent variation measure and the behavior of the implied expenditure shares. Appendix E incorporates the consumption-goods price wedge into the benchmark model of Section 2 of the paper and derives the corresponding equilibrium conditions.

Equation, proposition and assumption numbers without the “OA” prefix refer to the paper.

A Proof of Propositions

A.1 Proof of Proposition 1

Proposition 1. *Under Assumptions 1 to 4, along the ABGP of the structural transformation economy, the growth rate of real GDP, as measured by the Divisia index, declines over time.*

PROOF: The Divisia quantity index along the ABGP of the HRV structural transformation economy is

$$g_t^D = s_e(s_{g,t} g_{c_g} + (1 - s_{g,t}) g_{c_s,t}) + (1 - s_e) g_x.$$

The corresponding Divisia price index is

$$\pi_t^D = s_e(s_{g,t} \pi_g + (1 - s_{g,t}) \pi_s).$$

Since the investment good is the numeraire, $\pi_x = 0$. It is easy to show that $g_t^D = g_k - \pi_t^D$, where g_k is the growth rate of nominal income, as measured in units of the investment good. Since $s_{g,t}$ declines along the ABGP and $\pi_s > \pi_g$, consumption inflation increases, converging to service inflation asymptotically. It follows that, the growth rate of real GDP declines over time along the ABGP. $\square$

Quantitative Evaluation. Under our estimated and calibrated parameter values, the difference in growth rates between 1980 and 2023 is

$$g_{23}^D - g_{80}^D = \pi_{80}^D - \pi_{23}^D = s_e(s_{g,80} - s_{g,23})(\pi_g - \pi_s) < 0.$$

Under our calibration, $s_e \simeq 0.73$, $s_{g,23} \simeq 0.10$, $s_{g,80} \simeq 0.24$, and $\pi_s - \pi_g \simeq 0.02$, the model implies that the annual growth rate of real GDP in 2023 is 0.21 percentage points lower than in 1980.

A.2 Proof of Proposition 2

Proposition 2. *Under Assumptions 1–4, the indirect utility and expenditure functions associated with the Bellman representation of preferences in equation (14) are, respectively, given by*

$$u(m, P_g, P_s; \nu) = V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right) + \nu\left(m - (\nu P_s^\chi)^{\frac{1}{\chi-1}}\right), \quad (16)$$

and

$$f(w, P_g, P_s; \nu) = (\nu P_s^\chi)^{\frac{1}{\chi-1}} + \frac{w}{\nu} - \frac{V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right)}{\nu}, \quad (17)$$

where w is an arbitrary level of utility as in (14).

PROOF: In what follows we omit the index t to simplify notation. The household's primal problem of maximizing (14), subject to the budget constraint (15), can be solved in two stages. In the first stage, by the definition of the indirect utility function, the household solves

$$V(e, P_g, P_s) = \max_{\{c_g, c_s\}: P_g c_g + P_s c_s = e} U(c_g, c_s).$$

In the second stage, the household chooses x to solve

$$\max_x V(m - x, P_g, P_s) + \nu x.$$

The first-order condition is

$$\frac{\partial V}{\partial e}(e, P_g, P_s) = \nu.$$

Using the PIGL preferences in (6), $\frac{\partial V}{\partial e} = e^{\chi-1} P_s^{-\chi}$, we obtain

$$e^{\chi-1} P_s^{-\chi} = \nu \quad \Rightarrow \quad e = (\nu P_s^\chi)^{\frac{1}{\chi-1}}.$$

It follows that

$$x = m - e = m - (\nu P_s^\chi)^{\frac{1}{\chi-1}}.$$

Substituting back into (6), the indirect utility function is

$$u(m, P_g, P_s; \nu) = V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right) + \nu \left(m - (\nu P_s^\chi)^{\frac{1}{\chi-1}}\right).$$

The expenditure function is derived from the dual problem

$$\min_{e, x} e + x \quad \text{s.t.} \quad V(e, P_g, P_s) + \nu x = w,$$

where w is an arbitrary level of utility. The F.O.C.s with respect to e and x are

$$1 = \mu \frac{\partial V}{\partial e} = \mu e^{\chi-1} P_s^{-\chi} \quad \text{and} \quad 1 = \mu \nu,$$

where μ is the Lagrange multiplier associated with the constraint. Equating multipliers yields

$$e^{\chi-1} P_s^{-\chi} = \nu \quad \Rightarrow \quad e = (\nu P_s^\chi)^{\frac{1}{\chi-1}}.$$

The same condition as in the primal problem. Solving for x

$$x = \frac{w - V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right)}{\nu}.$$

Thus, the expenditure function is

$$f(w, P_g, P_s; \nu) = (\nu P_s^\chi)^{\frac{1}{\chi-1}} + \frac{w - V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right)}{\nu},$$

which completes the proof. $\square$

A.3 Proof of Proposition 3.

Proposition 3. *Under the conditions of Proposition 2:*

$$\hat{m}_{t,z,\tau} = m_\tau + B_{t,z,\tau} \hat{e}_{t,z},$$

where

$$B_{t,z,\tau} := \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,\tau}}{\gamma}\right) \left(\frac{P_{s,z}}{P_{s,\tau}}\right)^{\frac{\chi}{1-\chi}} - \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,z}}{\gamma}\right),$$

and, for $x \in \{z, \tau\}$,

$$\hat{s}_{g,t,x} = \eta \left(\frac{\hat{e}_{t,x}}{P_{s,x}}\right)^{-\chi} \left(\frac{P_{g,x}}{P_{s,x}}\right)^\gamma \quad \text{and} \quad \hat{e}_{t,x} = (\nu_t P_{s,x}^\chi)^{\frac{1}{\chi-1}}.$$

PROOF:

Step 1. From (21),

$$\hat{m}_{t,z,\tau} = f\left(u(m_\tau, P_{g,\tau}, P_{s,\tau}; \nu_t), P_{g,z}, P_{s,z}; \nu_t\right), \quad (21)$$

where $\hat{m}_{t,z,\tau}$ captures the income needed in period τ to attain the utility level of period τ under the preferences of period t and the prices of period z .

From Proposition 2,

$$u(m, P_g, P_s; \nu) = V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right) + \nu \left(m - (\nu P_s^\chi)^{\frac{1}{\chi-1}}\right), \quad (16)$$

and

$$f(w, P_g, P_s; \nu) = (\nu P_s^\chi)^{\frac{1}{\chi-1}} + \frac{w}{\nu} - \frac{V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right)}{\nu}. \quad (17)$$

Moreover, from Assumption 2,

$$V(e, P_g, P_s) = \frac{1}{\chi} \left(\frac{e}{P_s} \right)^\chi - \frac{\eta}{\gamma} \left(\frac{P_g}{P_s} \right)^\gamma - \frac{1}{\chi} + \frac{\eta}{\gamma}, \quad (6)$$

where $\eta > 0$ and $1 > \gamma > \chi > 0$.

Substituting (16) into (17), and the resulting expression into (21), we get

$$\begin{aligned} \hat{m}_{t,z,\tau} &= f\left(V(\hat{e}_{t,\tau}, P_{g,\tau}, P_{s,\tau}) + \nu_t(m_\tau - \hat{e}_{t,\tau}), P_{g,z}, P_{s,z}; \nu_t\right) \\ &= \hat{e}_{t,z} + \frac{1}{\nu_t} \left(V(\hat{e}_{t,\tau}, P_{g,\tau}, P_{s,\tau}) + \nu_t(m_\tau - \hat{e}_{t,\tau}) \right) - \frac{1}{\nu_t} V(\hat{e}_{t,z}, P_{g,z}, P_{s,z}), \end{aligned}$$

where $\hat{e}_{t,x} = (\nu_t P_{s,x}^\chi)^{\frac{1}{\chi-1}}$ for $x \in \{z, \tau\}$.

Rearranging terms, we get

$$\hat{m}_{t,z,\tau} = m_\tau + \left(\frac{V(\hat{e}_{t,\tau}, P_{g,\tau}, P_{s,\tau}) - \nu_t \hat{e}_{t,\tau}}{\nu_t} \right) - \left(\frac{V(\hat{e}_{t,z}, P_{g,z}, P_{s,z}) - \nu_t \hat{e}_{t,z}}{\nu_t} \right). \quad (22)$$

Step 2. Now evaluate the numerator of both terms in brackets. Using (6), for an arbitrary $x \in \{z, \tau\}$,

$$V(\hat{e}_{t,x}, P_{g,x}, P_{s,x}) - \nu_t \hat{e}_{t,x} = \frac{1}{\chi} \left(\frac{\hat{e}_{t,x}}{P_{s,x}} \right)^\chi - \frac{\eta}{\gamma} \left(\frac{P_{g,x}}{P_{s,x}} \right)^\gamma - \frac{1}{\chi} + \frac{\eta}{\gamma} - \nu_t \hat{e}_{t,x}.$$

From the definition $\hat{e}_{t,x} = (\nu_t P_{s,x}^\chi)^{\frac{1}{\chi-1}}$, we compute:

$$\frac{\hat{e}_{t,x}}{P_{s,x}} = (\nu_t P_{s,x})^{\frac{1}{\chi-1}}, \quad \left(\frac{\hat{e}_{t,x}}{P_{s,x}} \right)^\chi = (\nu_t P_{s,x})^{\frac{\chi}{\chi-1}} = \nu_t \hat{e}_{t,x}.$$

Thus

$$\frac{1}{\chi} \left(\frac{\hat{e}_{t,x}}{P_{s,x}} \right)^\chi - \nu_t \hat{e}_{t,x} = \left(\frac{1}{\chi} - 1 \right) \nu_t \hat{e}_{t,x}.$$

Hence:

$$V(\hat{e}_{t,x}, P_{g,x}, P_{s,x}) - \nu_t \hat{e}_{t,x} = \left(\frac{1}{\chi} - 1 \right) \nu_t \hat{e}_{t,x} - \frac{\eta}{\gamma} \left(\frac{P_{g,x}}{P_{s,x}} \right)^\gamma - \frac{1}{\chi} + \frac{\eta}{\gamma}.$$

Dividing by ν_t :

$$\frac{V(\hat{e}_{t,x}, P_{g,x}, P_{s,x}) - \nu_t \hat{e}_{t,x}}{\nu_t} = \left(\frac{1}{\chi} - 1\right) \hat{e}_{t,x} - \frac{\eta}{\gamma \nu_t} \left(\frac{P_{g,x}}{P_{s,x}}\right)^\gamma - \frac{1}{\chi \nu_t} + \frac{\eta}{\gamma \nu_t}.$$

Substituting into (22) for $x = \tau$ and $x = z$, the constant terms $-\frac{1}{\chi \nu_t} + \frac{\eta}{\gamma \nu_t}$ cancel. Therefore:

$$\hat{m}_{t,z,\tau} = m_\tau + \left(\frac{1}{\chi} - 1\right) (\hat{e}_{t,\tau} - \hat{e}_{t,z}) - \frac{\eta}{\gamma \nu_t} \left(\left(\frac{P_{g,\tau}}{P_{s,\tau}}\right)^\gamma - \left(\frac{P_{g,z}}{P_{s,z}}\right)^\gamma \right). \quad (22')$$

Step 3. From (18), the equivalent variation measure $\hat{e}_{t,x} = (\nu_t P_{s,x}^\chi)^{\frac{1}{\chi-1}}$ implying

$$\frac{1}{\nu_t} \left(\frac{\hat{e}_{t,x}}{P_{s,x}}\right)^\chi = \hat{e}_{t,x}.$$

Use now Roy's identity (7) for the equivalent variation measure $\hat{s}_{g,t,x}$, $x \in \{z, \tau\}$,

$$\hat{s}_{g,t,x} = \eta \left(\frac{\hat{e}_{t,x}}{P_{s,x}}\right)^{-\chi} \left(\frac{P_{g,x}}{P_{s,x}}\right)^\gamma \Rightarrow \frac{\eta}{\gamma \nu_t} \left(\frac{P_{g,x}}{P_{s,x}}\right)^\gamma = \frac{1}{\gamma} \hat{s}_{g,t,x} \hat{e}_{t,x}.$$

Substituting it into (22') and using $\hat{e}_{t,\tau} = \hat{e}_{t,z} \left(\frac{P_{s,\tau}}{P_{s,z}}\right)^{\frac{\chi}{\chi-1}}$, we obtain

$$\hat{m}_{t,z,\tau} = m_\tau + B_{t,z,\tau} \hat{e}_{t,z},$$

where

$$B_{t,z,\tau} := \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,\tau}}{\gamma}\right) \left(\frac{P_{s,z}}{P_{s,\tau}}\right)^{\frac{\chi}{1-\chi}} - \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,z}}{\gamma}\right),$$

which completes the proof. $\square$

A.4 Proof of Proposition 4.

Proposition 4. *Under the conditions of Proposition 2*

$$\frac{d \log \hat{m}_{t,z,\tau}}{d\tau} = \frac{m_\tau}{\hat{m}_{t,z,\tau}} \cdot (g_\tau^D + \text{dev}_{t,z,\tau}) := \hat{g}_{t,z,\tau}^D,$$

where

$$g_\tau^D = s_{e,\tau} (s_{g,\tau} g_{c_g,\tau} + (1 - s_{g,\tau}) g_{c_s,\tau}) + (1 - s_{e,\tau}) g_{x,\tau},$$

and

$$\text{dev}_{t,z,\tau} = \left(s_{e,\tau} s_{g,\tau} - v_{z,\tau} \check{s}_{e,t,z,\tau} \hat{s}_{g,t,\tau} \right) \pi_g + \left(s_{e,\tau} (1 - s_{g,\tau}) - v_{z,\tau} \check{s}_{e,t,z,\tau} (1 - \hat{s}_{g,t,\tau}) \right) \pi_s,$$

with

$$\check{s}_{e,t,z,\tau} := \frac{\hat{e}_{t,z}}{m_\tau} \quad \text{and} \quad v_{z,\tau} := \left(\frac{P_{s,z}}{P_{s,\tau}} \right)^{\frac{\chi}{1-\chi}}.$$

PROOF:

Step 1. From Proposition 3,

$$\hat{m}_{t,z,\tau} = m_\tau + B_{t,z,\tau} \hat{e}_{t,z},$$

where

$$B_{t,z,\tau} := \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,\tau}}{\gamma} \right) \left(\frac{P_{s,z}}{P_{s,\tau}} \right)^{\frac{\chi}{1-\chi}} - \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,z}}{\gamma} \right),$$

Differentiating $B_{t,z,\tau}$ with respect to τ ,

$$\frac{dB_{t,z,\tau}}{d\tau} = -\frac{v_{z,\tau}}{\gamma} \frac{d\hat{s}_{g,t,\tau}}{d\tau} + \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,\tau}}{\gamma} \right) \frac{dv_{z,\tau}}{d\tau},$$

where $v_{z,\tau} = \left(\frac{P_{s,z}}{P_{s,\tau}} \right)^{\frac{\chi}{1-\chi}}$. From the definition of $v_{z,\tau}$ above,

$$\frac{dv_{z,\tau}}{d\tau} = -\frac{\chi}{1-\chi} \pi_s v_{z,\tau}.$$

From the definition of $\hat{s}_{g,t,x}$ and $\hat{e}_{t,x}$, we get

$$\hat{s}_{g,t,\tau} = \eta \left(\frac{\hat{e}_{t,\tau}}{P_{s,\tau}} \right)^{-\chi} \left(\frac{P_{g,\tau}}{P_{s,\tau}} \right)^\gamma = \eta \nu_t^{\frac{\chi}{1-\chi}} P_{s,\tau}^{\frac{\chi}{1-\chi}-\gamma} P_{g,\tau}^\gamma.$$

Differentiating with respect to τ ,

$$\frac{d\hat{s}_{g,t,\tau}}{d\tau} = \hat{s}_{g,t,\tau} \left[\left(\frac{\chi}{1-\chi} - \gamma \right) \pi_s + \gamma \pi_g \right].$$

Plugging both derivatives into $\frac{dB_{t,z,\tau}}{d\tau}$, we get

$$\frac{dB_{t,z,\tau}}{d\tau} = -v_{z,\tau} \left(\frac{\hat{s}_{g,t,\tau}}{\gamma} \left[\left(\frac{\chi}{1-\chi} - \gamma \right) \pi_s + \gamma \pi_g \right] + \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,\tau}}{\gamma} \right) \frac{\chi}{1-\chi} \pi_s \right),$$

which collapses to

$$\frac{dB_{t,z,\tau}}{d\tau} = -v_{z,\tau} \left(\hat{s}_{g,t,\tau} \pi_g + (1 - \hat{s}_{g,t,\tau}) \pi_s \right).$$

Step 2. The derivative of $\log \hat{m}_{t,z,\tau}$ with respect to τ is

$$\frac{d \log \hat{m}_{t,z,\tau}}{d\tau} = \frac{\dot{m}_\tau}{\hat{m}_{t,z,\tau}} + \frac{dB_{t,z,\tau}}{d\tau} \cdot \frac{\hat{e}_{t,z}}{\hat{m}_{t,z,\tau}}.$$

Since

$$\frac{\dot{m}_\tau}{m_\tau} = s_{e,\tau} g_{e,\tau} + (1 - s_{e,\tau}) g_{x,\tau},$$

and

$$g_{e,\tau} = s_{g,\tau} (g_{c_g,\tau} + \pi_g) + (1 - s_{g,\tau}) (g_{c_s,\tau} + \pi_s),$$

Then,

$$\frac{d \log \hat{m}_{t,z,\tau}}{d\tau} = \frac{m_\tau}{\hat{m}_{t,z,\tau}} \cdot (g_\tau^D + \text{dev}_{t,z,\tau}) := \hat{g}_{t,z,\tau}^D,$$

where

$$g_\tau^D = s_{e,\tau} (s_{g,\tau} g_{c_g,\tau} + (1 - s_{g,\tau}) g_{c_s,\tau}) + (1 - s_{e,\tau}) g_{x,\tau},$$

and

$$\text{dev}_{t,z,\tau} = \left(s_{e,\tau} s_{g,\tau} - v_{z,\tau} \check{s}_{e,t,z,\tau} \hat{s}_{g,t,\tau} \right) \pi_g + \left(s_{e,\tau} (1 - s_{g,\tau}) - v_{z,\tau} \check{s}_{e,t,z,\tau} (1 - \hat{s}_{g,t,\tau}) \right) \pi_s,$$

with

$$\check{s}_{e,t,z,\tau} := \frac{\hat{e}_{t,z}}{m_\tau},$$

which completes the proof. $\square$

A.5 Proof of Proposition 5

Proposition 5. *The instantaneous growth rates of the current-base and reference-base Fisher–Shell indices, when evaluated at the base time, are equal to the corresponding Divisia index:*

$$\left. \frac{d\mathcal{P}_{t_1,z}}{dz} \right|_{z=t_1} = g_{t_1}^D \quad \text{and} \quad \left. \frac{d\mathcal{L}_{t_0,z}}{dz} \right|_{z=t_0} = g_{t_0}^D. \quad (25)$$

Moreover, for all $z \in (t_0, t_1)$, $t_0 < t_1$, conditional on s_{g,t_0} being small enough,

$$\frac{d\mathcal{P}_{t_1,z}}{dz} < g_z^D < \frac{d\mathcal{L}_{t_0,z}}{dz}, \quad (26)$$

and therefore

$$\mathcal{L}_{t_0,z} > \mathcal{D}_z > \mathcal{P}_{t_1,z}.$$

PROOF:

Part I. Let us first prove (25). From the definition of the current-base and reference-base Fisher–Shell indices in (23) and (24), respectively,

$$\mathcal{P}_{t_1,z} = \log \hat{m}_{t_1,t_1,z} - \log \hat{m}_{t_1,t_1,t_0} \quad \text{and} \quad \mathcal{L}_{t_0,z} = \log \hat{m}_{t_0,t_0,z} - \log m_{t_0}.$$

Then, the growth rate of the current-base and reference-base Fisher–Shell indices at current and reference times, respectively, are

$$\left. \frac{d\mathcal{P}_{t_1,z}}{dz} \right|_{z=t_1} = \frac{1}{\hat{m}_{t_1,t_1,z}} \cdot \left. \frac{d\hat{m}_{t_1,t_1,z}}{dz} \right|_{z=t_1} \quad \text{and} \quad \left. \frac{d\mathcal{L}_{t_0,z}}{dz} \right|_{z=t_0} = \frac{1}{\hat{m}_{t_0,t_0,z}} \cdot \left. \frac{d\hat{m}_{t_0,t_0,z}}{dz} \right|_{z=t_0}.$$

From Proposition 4,

$$\left. \frac{d \log \hat{m}_{t,t,z}}{dz} \right|_{z=t} = g_t^D$$

since

$$\hat{m}_{t,t,t} = m_t \quad \text{and} \quad \text{dev}_{t,t,t} = 0.$$

The previous result is true, because $v_{t,t} = 1$, $\check{s}_{e,t,t} = s_{e,t}$ and $\hat{s}_{g,t,t} = s_{g,t}$. We have then shown that

$$\left. \frac{d\mathcal{P}_{t_1,z}}{dz} \right|_{z=t_1} = g_{t_1}^D \quad \text{and} \quad \left. \frac{d\mathcal{L}_{t_0,z}}{dz} \right|_{z=t_0} = g_{t_0}^D.$$

Part II. Let us now prove (26). From the definition of $\mathcal{P}_{t_1,z}$ and $\mathcal{L}_{t_0,z}$,

$$\frac{d\mathcal{P}_{t_1,z}}{dz} = \frac{d \log \hat{m}_{t_1,t_1,z}}{dz} \quad \text{and} \quad \frac{d\mathcal{L}_{t_0,z}}{dz} = \frac{d \log \hat{m}_{t_0,t_0,z}}{dz},$$

while from Proposition 4

$$\frac{d \log \hat{m}_{t,t,z}}{dz} = \frac{m_z}{\hat{m}_{t,t,z}} \cdot (g_z^D + \text{dev}_{t,t,z}),$$

where

$$\text{dev}_{t,t,z} = \left(s_{e,z} s_{g,z} - v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} \right) (\pi_g - \pi_s) + \left(s_{e,z} - v_{t,z} \check{s}_{e,t,t,z} \right) \pi_s,$$

with

$$s_{e,z} = \frac{e_z}{m_z},$$

$$s_{g,z} = \eta \left(\frac{e_z}{P_{s,z}} \right)^{-\chi} \left(\frac{P_{g,z}}{P_{s,z}} \right)^{\gamma},$$

$$v_{t,z} = \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{1-\chi}},$$

$$\check{s}_{e,t,z} = \frac{e_t}{m_z},$$

$$\hat{s}_{g,t,z} = \eta \left(\frac{\hat{e}_{t,z}}{P_{s,z}} \right)^{-\chi} \left(\frac{P_{g,z}}{P_{s,z}} \right)^{\gamma},$$

$$\hat{e}_{t,z} = e_t \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{\chi-1}}.$$

All of them depend exclusively on equilibrium objects.

Step 1: Let's study the factor $\frac{m_z}{\hat{m}_{t,t,z}}$. We show below that

$$\text{if } s_{g,t_0} < \min_{z \in (t_0, t_1)} \frac{\pi_s}{\pi_s - \pi_g} e^{\left(\frac{\chi}{1-\chi} \pi_s - \gamma(\pi_s - \pi_g) \right)(t-z)} \quad \Rightarrow \quad \frac{m_z}{\hat{m}_{t_0,t_0,z}} > 1 > \frac{m_z}{\hat{m}_{t_1,t_1,z}}.$$

From Proposition 3

$$\hat{m}_{t,t,z} = m_z + B_{t,t,z} e_t,$$

where

$$\begin{aligned} B_{t,t,z} &= \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,z}}{\gamma} \right) \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{1-\chi}} - \left(\frac{1}{\chi} - 1 - \frac{s_{g,t}}{\gamma} \right) \\ &= \left(\frac{1}{\chi} - 1 \right) \left(\left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{1-\chi}} - 1 \right) + \frac{s_{g,t}}{\gamma} \left(1 - \left(\frac{P_{g,z}/P_{s,z}}{P_{g,t}/P_{s,t}} \right)^{\gamma} \right). \end{aligned}$$

The last equality is true since

$$s_{g,t} - \hat{s}_{g,t,z} \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{1-\chi}} = s_{g,t} \left(1 - \left(\frac{P_{g,z}/P_{s,z}}{P_{g,t}/P_{s,t}} \right)^{\gamma} \right).$$

At equilibrium

$$P_{g,z} = e^{\pi_g(z-t_0)} \quad \text{and} \quad P_{s,z} = e^{\pi_s(z-t_0)},$$

with $\pi_s > 0$, $\pi_s > \pi_g$ and $1 > \gamma > \chi > 0$. Then,

$$B_{t,t,z} = \left(\frac{1}{\chi} - 1 \right) \left(e^{\frac{\chi}{1-\chi} \pi_s(t-z)} - 1 \right) + \frac{s_{g,t}}{\gamma} \left(1 - e^{\gamma(\pi_s - \pi_g)(t-z)} \right),$$

with $B_{t_1,t_1,t_1} = B_{t_0,t_0,t_0} = 0$.

Differentiate it with respect to z :

$$\frac{dB_{t,t,z}}{dz} = -\pi_s e^{\frac{\chi}{1-\chi}\pi_s(t-z)} + s_{g,t}(\pi_s - \pi_g) e^{\gamma(\pi_s - \pi_g)(t-z)}.$$

Since $\pi_s > 0$, $\pi_s > \pi_g$ and $s_{g,t} \in (0, 1)$, the first term is negative and the second is positive.

For all $z \in (t_0, t_1)$, $\frac{dB_{t_0,t_0,z}}{dz} < 0$ if and only if

$$s_{g,t_0} < \min_{z \in (t_0, t_1)} \frac{\pi_s}{\pi_s - \pi_g} e^{\left(\frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g)\right)(t_0 - z)}, \quad (s_{g,t_0})$$

Under this condition, for all $z \in (t_0, t_1)$, $B_{t_0,t_0,z} < 0$.

Notice that if $\frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g) < 0$, the condition becomes $s_{g,t_0} < \frac{\pi_s}{\pi_s - \pi_g}$. Otherwise $s_{g,t_0} < \frac{\pi_s}{\pi_s - \pi_g} e^{\left(\frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g)\right)(t_0 - t_1)} < \frac{\pi_s}{\pi_s - \pi_g}$.

The consumption share of good g at time t , where $e_t = e_{t_0} e^{g_e(t-t_0)}$, is

$$s_{g,t} = \eta \left(\frac{e_t}{P_{s,t}} \right)^{-\chi} \left(\frac{P_{g,t}}{P_{s,t}} \right)^{\gamma} = s_{g,t_0} e^{E(t-t_0)},$$

with $s_{g,t_0} = \eta e_{t_0}^{-\chi}$ and $E = -\chi(g_e - \pi_s) - \gamma(\pi_s - \pi_g)$. Given $g_e > \pi_s$, $\pi_s > \pi_g$ and $1 > \gamma > \chi > 0$, $E < 0$, so $s_{g,t}$ is strictly decreasing over time.

For all $z \in (t_0, t_1)$, $\frac{dB_{t_1,t_1,z}}{dz} < 0$ if and only if

$$s_{g,t_1} < \min_{z \in (t_0, t_1)} \frac{\pi_s}{\pi_s - \pi_g} e^{\left(\frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g)\right)(t_1 - z)}, \quad (s_{g,t_1})$$

Under this condition, for all $z \in (t_0, t_1)$, $B_{t_1,t_1,z} > 0$. Notice that this condition can be written as

$$s_{g,t_0} e^{E(t_1-t_0)} < e^{\left(\frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g)\right)(t_1-t_0)} \min_{z \in (t_0, t_1)} \frac{\pi_s}{\pi_s - \pi_g} e^{\left(\frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g)\right)(t_0 - z)},$$

with e_{t_0} determined in (12). Since $E < \frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g)$, condition (s_{g,t_0}) implies (s_{g,t_1}) .

Step 2: Sign of $\text{dev}_{t,t,z}$. Let's study the sign of

$$\text{dev}_{t,t,z} = \left(s_{e,z} s_{g,z} - v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} \right) (\pi_g - \pi_s) + \left(s_{e,z} - v_{t,z} \check{s}_{e,t,t,z} \right) \pi_s.$$

With this in mind, we study first the sign of both parenthesis at the right-hand-side.

(A) Let us show that the first term in the equation above

$$s_{e,z} s_{g,z} - v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} > 0 \iff \nu_t > \nu_z,$$

and negative iff $\nu_t < \nu_z$. Let's prove it by steps.

Write explicitly each term at the left-hand-side of the inequality above:

$$s_{e,z} s_{g,z} = \frac{e_z}{m_z} \cdot \eta \left(\frac{e_z}{P_{s,z}} \right)^{-\chi} \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma$$

and

$$v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} = \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{1-\chi}} \cdot \frac{e_t}{m_z} \cdot \eta \left(\frac{\hat{e}_{t,z}}{P_{s,z}} \right)^{-\chi} \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma.$$

Since, $\hat{e}_{t,z} = e_t \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{\chi-1}}$,

$$v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} = \frac{e_t}{m_z} \cdot \eta \left(\frac{e_t}{P_{s,t}} \right)^{-\chi} \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma.$$

Take the common factor $\frac{\eta}{m_z} \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma$:

$$s_{e,z} s_{g,z} - v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} = \frac{\eta}{m_z} \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma (e_z^{1-\chi} P_{s,z}^\chi - e_t^{1-\chi} P_{s,t}^\chi).$$

From (18),

$$e_t = (\nu_t P_{s,t}^\chi)^{\frac{1}{\chi-1}} \quad \Rightarrow \quad e_t \nu_t^{\frac{-\chi}{\chi-1}} P_{s,t}^{\frac{\chi}{1-\chi}} = \nu_t^{-1}.$$

Hence, using the fact that

$$e_t^{1-\chi} P_{s,t}^\chi = \frac{1}{\nu_t} \quad \text{and} \quad e_z^{1-\chi} P_{s,z}^\chi = \frac{1}{\nu_z},$$

we obtain

$$s_{e,z} s_{g,z} - v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} = \frac{\eta}{m_z} \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma \left(\frac{1}{\nu_z} - \frac{1}{\nu_t} \right).$$

Since $\eta > 0$, $m_z > 0$, $P_{g,z} > 0$ and $P_{s,z} > 0$, the sign is determined by $\frac{1}{\nu_z} - \frac{1}{\nu_t}$. Therefore

$$s_{e,z} s_{g,z} - v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} > 0 \quad \Longleftrightarrow \quad \frac{1}{\nu_z} > \frac{1}{\nu_t} \quad \Longleftrightarrow \quad \nu_t > \nu_z,$$

and it is negative if and only if $\nu_t < \nu_z$.

Since at equilibrium $\pi_s > \pi_g$, then

$$\left(s_{e,z} s_{g,z} - v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} \right) (\pi_g - \pi_s) < 0 \quad \Longleftrightarrow \quad \frac{1}{\nu_z} > \frac{1}{\nu_t} \quad \Longleftrightarrow \quad \nu_t > \nu_z,$$

and it is positive if and only if $\nu_t < \nu_z$.

(B) Let us now show that

$$s_{e,z} - v_{t,z} \check{s}_{e,t,t,z} > 0 \iff \nu_t > \nu_z,$$

and negative if $\nu_t < \nu_z$.

Since:

$$s_{e,z} = \frac{e_z}{m_z}, \quad v_{t,z} = \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{1-\chi}} \quad \text{and} \quad \check{s}_{e,t,t,z} = \frac{e_t}{m_z}.$$

Thus:

$$s_{e,z} - v_{t,z} \check{s}_{e,t,t,z} = \frac{e_z}{m_z} - \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{1-\chi}} \cdot \frac{e_t}{m_z}.$$

Take common factor $\frac{P_{s,z}^{\frac{\chi}{1-\chi}}}{m_z}$:

$$= \frac{P_{s,z}^{\frac{\chi}{1-\chi}}}{m_z} \left(e_z P_{s,z}^{\frac{\chi}{1-\chi}} - e_t P_{s,t}^{\frac{\chi}{1-\chi}} \right).$$

Since from (18)

$$e_t = (\nu_t P_{s,t}^\chi)^{\frac{1}{\chi-1}} \implies e_t P_{s,t}^{\frac{\chi}{1-\chi}} = \nu_t^{\frac{1}{\chi-1}}$$

Thus:

$$s_{e,z} - v_{t,z} \check{s}_{e,t,t,z} = \frac{P_{s,z}^{\frac{\chi}{1-\chi}}}{m_z} \left(\nu_z^{\frac{1}{\chi-1}} - \nu_t^{\frac{1}{\chi-1}} \right)$$

Since $\frac{1}{\chi-1} < 0$, because $\chi \in (0, 1)$, the function $x^{\frac{1}{\chi-1}}$ is decreasing in x . Hence:

$$\nu_z^{\frac{1}{\chi-1}} - \nu_t^{\frac{1}{\chi-1}} > 0 \iff \nu_z < \nu_t.$$

Consequently:

$$s_{e,z} - v_{t,z} \check{s}_{e,t,t,z} > 0 \iff \nu_t > \nu_z.$$

From the inequalities above, since $\pi_s > 0$ and $\pi_s > \pi_g$ at equilibrium, for all $z \in (t_0, t_1)$, $\text{dev}_{t_0,t_0,z} > 0$ and $\text{dev}_{t_1,t_1,z} < 0$.

Step 3. From the analysis above, for the reference-base t_0 , for all $z \in (t_0, t_1)$, $\frac{m_z}{\hat{m}_{t_0,t_0,z}} > 1$ and $\text{dev}_{t_0,t_0,z} > 0$ implying that $\frac{d\mathcal{L}_{t_0,z}}{dz} = \frac{d \log \hat{m}_{t_0,t_0,z}}{dz} > g_z^D$ and $\mathcal{L}_{t_0,z} > \mathcal{D}_z$. Using a similar argument, $\frac{d\mathcal{P}_{t_1,z}}{dz} = \frac{d \log \hat{m}_{t_1,t_1,z}}{dz} < g_z^D$ and $\mathcal{P}_{t_1,z} < \mathcal{D}_z$. $\square$

B Boppart's 2014 Lemma 1

Lemma 1 in Boppart (2014) provides sufficient conditions for the indirect utility function $V(e, P_g, P_s)$ to be well-defined. In particular, for all $t \in (t_0, t_1)$, it requires

$$e_t^\chi \geq \frac{1-\chi}{1-\gamma} \eta P_{g,t}^\gamma P_{s,t}^{\chi-\gamma}.$$

Under the calibration presented in Section 4, this condition holds along the equilibrium path of the structural transformation economy. Using (7), the inequality above is equivalent to

$$s_{g,t} \leq \frac{1-\gamma}{1-\chi}.$$

Given the calibrated values of $\gamma = 0.714$ and $\chi = 0.360$, this bound implies that $s_{g,t} \leq 0.447$. In our simulated economy, this holds for all $t \geq 1980$ along the ABGP, as shown in Figure 6.

Moreover, for the preference order behind a current-base Fisher–Shell index $\mathcal{P}_{t_1,z}$, $z \in (t_0, t_1)$, to be well-defined,

$$\hat{e}_{t_1,z}^\chi \geq \frac{1-\chi}{1-\gamma} \eta P_{g,z}^\gamma P_{s,z}^{\chi-\gamma} \quad \text{or equivalently} \quad \hat{s}_{g,t_1,z} \leq \frac{1-\gamma}{1-\chi}.$$

and

$$e_{t_1}^\chi \geq \frac{1-\chi}{1-\gamma} \eta P_{g,t_1}^\gamma P_{s,t_1}^{\chi-\gamma}. \quad \text{or equivalently} \quad s_{g,t_1} \leq \frac{1-\gamma}{1-\chi}.$$

The first condition corresponds to the application of Boppart's Lemma 1 to the indirect utility $u(\hat{e}_{t_1,z}, P_{g,z}, P_{s,z}; \nu_{t_1})$ and the second to expenditure function $f(w, P_{g,t_1}, P_{s,t_1}; \nu_{t_1})$. The second condition holds, since Boppart's Lemma 1 holds at any point in the equilibrium path. Since ν_z is decreasing along the equilibrium path,

$$\hat{e}_{t_1,z} = \left(\frac{\nu_{t_1}}{\nu_z} \right)^{\frac{1}{\chi-1}} e_z > e_z > \frac{1-\chi}{1-\gamma} \eta P_{g,z}^\gamma P_{s,z}^{\chi-\gamma},$$

for all $z \in (t_0, t_1)$ at the ABGP equilibrium of the structural transformation economy. This can be observed in the behavior of $\hat{s}_{g,t_1,z}$ in Figure 6, which is always below the critical value 0.447.

For the preference order behind a reference-base Fisher–Shell index $\mathcal{L}_{t_0,z}$, $z \in (t_0, t_1)$, to be well-defined,

$$\hat{e}_{t_0,z}^\chi \geq \frac{1-\chi}{1-\gamma} \eta P_{g,z}^\gamma P_{s,z}^{\chi-\gamma} \quad \text{or equivalently} \quad \hat{s}_{g,t_0,z} \leq \frac{1-\gamma}{1-\chi}.$$

and

$$e_{t_0}^\chi \geq \frac{1-\chi}{1-\gamma} \eta P_{g,t_1}^\gamma P_{s,t_1}^{\chi-\gamma}. \quad \text{or equivalently} \quad s_{g,t_0} \leq \frac{1-\gamma}{1-\chi}.$$

Figure 6 shows that $\hat{s}_{g,t_0,z}$ is also always below the critical value of 0.447, guaranteeing that both conditions are satisfied.

Hence, the regularity condition of Lemma 1 in Boppart (2014) holds along the equilibrium path. As a result, the baseline indirect utility function, as well as the preference orderings underlying the current-base and reference-base indices, are well-defined.

C Indices for Real Consumption Expenditure

This appendix applies the welfare measures discussed in this paper to an economy without capital, such as the purely endowment economy in Baqaee and Burstein (2022). As shown below, the fixed-base Fisher–Shell index continues to exhibit the same qualitative distortions—still quantitatively important—in an economy where the utility function is non-homothetic and income effects are time-varying.

The indirect utility function associated with the instantaneous utility function $U(c_{g,t}, c_{s,t})$ is

$$V(e_t, P_{g,t}, P_{s,t}) = \frac{1}{\chi} \left(\left(\frac{e_t}{P_{s,t}} \right)^\chi - 1 \right) + \frac{\eta}{\gamma} \left(1 - \left(\frac{P_{g,t}}{P_{s,t}} \right)^\gamma \right). \quad (6)$$

The associated expenditure function is

$$E(v_t, P_{g,t}, P_{s,t}) = \chi^{\frac{1}{\chi}} \left(v_t + \frac{1}{\chi} - \frac{\eta}{\gamma} \left(1 - \left(\frac{P_{g,t}}{P_{s,t}} \right)^\gamma \right) \right)^{\frac{1}{\chi}} P_{s,t}, \quad (\text{OA.1})$$

where v_t captures consumption utility level.

Let us compare welfare at time z with welfare at time t by means of the equivalent variation measure

$$\tilde{e}_{t,z} = E(V(e_z, P_{g,z}, P_{s,z}), P_{g,t}, P_{s,t}). \quad (\text{OA.2})$$

The representation of the instantaneous utility of consumption is time invariant. Notice that $\tilde{e}_{t,z}$ does not depend on the marginal value of capital ν_t , as is the case for $\hat{e}_{t,z}$ in (22).

Substituting (6) and (OA.1) into (OA.2), we get

$$\tilde{e}_{t,z} = \left(\left(\frac{e_z}{P_{s,z}} \right)^\chi + \frac{\eta\chi}{\gamma} \left(\left(\frac{P_{g,t}}{P_{s,t}} \right)^\gamma - \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma \right) \right)^{\frac{1}{\chi}} P_{s,t}.$$

Chained Fisher–Shell Index. In line with [Durán and Licandro \(2025\)](#), it is easy to show that a chained Divisia index is welfare based. Differentiate (OA.2) with respect to τ

$$\frac{d\tilde{e}_{z,\tau}}{d\tau} = \frac{dE(V_\tau, P_{g,z}, P_{s,z})}{dV_\tau} \times \frac{dV(e_\tau, P_{g,\tau}, P_{s,\tau})}{d\tau},$$

with

$$\frac{dE(V_\tau, P_{g,z}, P_{s,z})}{dV_\tau} = \chi^{\frac{1}{\chi}-1} \left(V_\tau + \frac{1}{\chi} - \frac{\eta}{\gamma} \left(1 - \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma \right) \right)^{\frac{1}{\chi}-1} P_{s,z} = \tilde{e}_{z,\tau} \left(\frac{\tilde{e}_{z,\tau}}{P_{s,z}} \right)^{-\chi}$$

and

$$\frac{dV(e_\tau, P_{g,\tau}, P_{s,\tau})}{d\tau} = \left(\frac{e_\tau}{P_{s,\tau}} \right)^\chi (g_e - \pi_s) - \eta \left(\frac{P_{g,\tau}}{P_{s,\tau}} \right)^\gamma (\pi_g - \pi_s),$$

where $g_e = \frac{\dot{e}_\tau}{e_\tau}$, $\pi_g = \frac{\dot{P}_{g,\tau}}{P_{g,\tau}}$ and $\pi_s = \frac{\dot{P}_{s,\tau}}{P_{s,\tau}}$. All three growth rates are time invariant in equilibrium. From (7)

$$\eta \left(\frac{P_{g,\tau}}{P_{s,\tau}} \right)^\gamma = s_{g,\tau} \left(\frac{e_\tau}{P_{s,\tau}} \right)^\chi.$$

Then,

$$\frac{dV(e_\tau, P_{g,\tau}, P_{s,\tau})}{d\tau} = \left(\frac{e_\tau}{P_{s,\tau}} \right)^\chi \left(g_e - s_{g,\tau} \pi_g - (1 - s_{g,\tau}) \pi_s \right).$$

From the definition of total consumption expenditure

$$g_e = s_{g,\tau} (g_{c_g} + \pi_g) + (1 - s_{g,\tau}) (g_{c_s,\tau} + \pi_s).$$

Consequently, the growth rate at time τ when evaluated at time- z prices is

$$\frac{d\tilde{e}_{z,\tau}}{d\tau} \frac{1}{\tilde{e}_{z,\tau}} = \left(s_{g,\tau} g_{c_g} + (1 - s_{g,\tau}) g_{c_s,\tau} \right) \left(\frac{e_\tau P_{s,z}}{\tilde{e}_{z,\tau} P_{s,\tau}} \right)^\chi.$$

When, for all $z \in (t_0, t_1)$, income around z is evaluated using time- z prices, the equivalent variation measure at $\tau = z$ is the Divisia index, i.e.,

$$\left. \frac{d \log \tilde{e}_{z,\tau}}{d\tau} \right|_{\tau=z} = g_{e,z}^D.$$

When normalized to zero at $z = t_0$, the chained Fisher–Shell index becomes the chained Divisia index

$$\mathcal{D}_{e,z} = \int_{t_0}^z g_{e,h}^D dh.$$

From the perspective of the chained Fisher–Shell index, the real growth rate of consumption expenditure at any point in time is computed using the prices prevailing at that moment. It captures cumulative welfare gains, as perceived by the representative agent over time. It is from this perspective that the chained Divisia index is welfare-based.

Fixed-Base Fisher–Shell Indices. The current-base Fisher–Shell index is

$$\mathcal{P}_{e,t_1,z} = \log(\tilde{e}_{t_1,z}) - \log(\tilde{e}_{t_1,t_0}).$$

In turn, the reference-base Fisher–Shell index is given by

$$\mathcal{L}_{e,t_0,z} = \log(\tilde{e}_{t_0,z}) - \log(\tilde{e}_{t_0,t_0}).$$

Following the derivations above, the growth rate at z of a fixed-base Fisher–Shell index for consumption expenditure in this context, with time- t base preferences and prices is

$$g_{e,t,z} = \frac{d \log \tilde{e}_{t,z}}{dz} = g_{e,z}^D \left(\frac{e_z}{\tilde{e}_{t,z}} \frac{P_{s,t}}{P_{s,z}} \right)^\chi.$$

Quantitative results. Under the same calibration as in the main text, the resulting consumption expenditure indices are presented in Figure [OA.1](#) below. As can be observed, the current-base FS index underestimates economic growth relative to the Divisia index, while the reference-base FS index overestimates it. As a result, the required permanent revision of the current-base FS index exhibits a declining growth pattern and successive downward revisions of past growth.

D Properties of the Equivalent Variation Measure

Starting from the equivalent variation measure defined in equation [\(21\)](#),

$$\hat{m}_{t,z,\tau} = f\left(u(m_\tau, P_{g,\tau}, P_{s,\tau}; \nu_t), P_{g,z}, P_{s,z}; \nu_t\right),$$

we now examine its structure and implications. This equivalent variation results from solving two consecutive problems.

First stage. At the first stage, the representative household uses time- t preferences to evaluate the utility attainable at time- τ with income m_τ and prices $P_{g,\tau}$ and $P_{s,\tau}$. To do so,

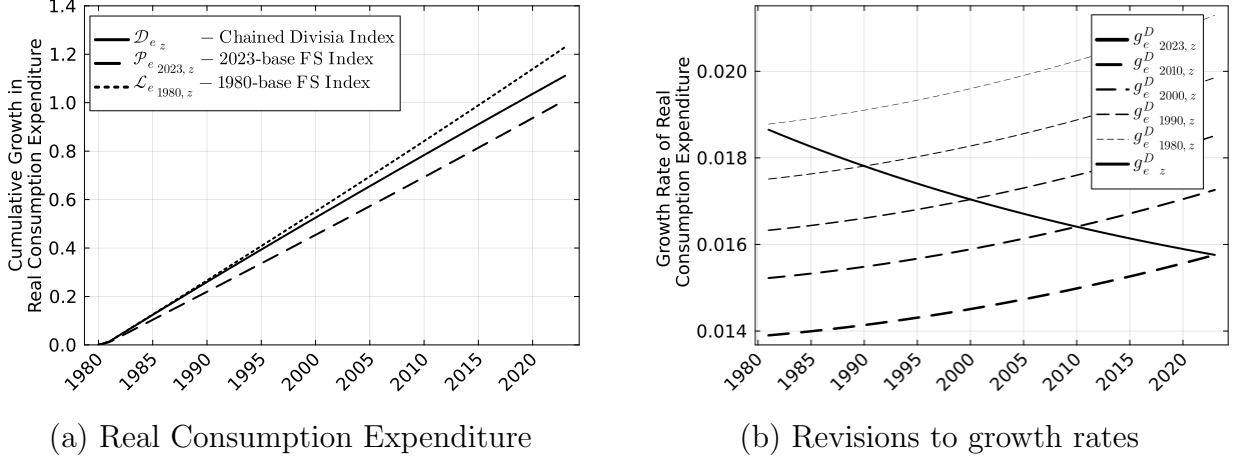


Figure OA.1: Real Consumption Expenditure Indices and Revisions to Growth Rates.

Note: Panel (a): Divisia (solid line), 1980-base Fisher–Shell (dotted) and 2023-base Fisher–Shell (dashed). Panel (b): Growth rates of fixed-base FS indices with bases 1980, 1990, 2000, 2010 and 2023.

the household solves the following hypothetical problem:

$$\max_{\tilde{c}_g, \tilde{c}_s, \tilde{x}} U(\tilde{c}_{g,t,\tau}, \tilde{c}_{s,t,\tau}) + \nu_t \tilde{x}_{t,\tau}$$

subject to

$$m_\tau = P_{g,\tau} \tilde{c}_{g,t,\tau} + P_{s,\tau} \tilde{c}_{s,t,\tau} + \tilde{x}_{t,\tau}.$$

Using the indirect utility function (6), this problem reduces to:

$$\max_{\tilde{e}, \tilde{x}} V(\tilde{e}_{t,\tau}, P_{g,\tau}, P_{s,\tau}) + \nu_t \tilde{x}_{t,\tau}$$

subject to

$$m_\tau = \tilde{e}_{t,\tau} + \tilde{x}_{t,\tau},$$

where $\tilde{e}_{t,\tau} = P_{g,\tau} \tilde{c}_{g,t,\tau} + P_{s,\tau} \tilde{c}_{s,t,\tau}$.

The first-order condition with respect to $\tilde{e}_{t,\tau}$ yields, after some algebra,

$$\tilde{e}_{t,\tau} = \left(\nu_t P_{s,\tau}^\chi \right)^{\frac{1}{\chi-1}}.$$

Since the price of services grows with τ , the hypothetical consumption expenditure $\tilde{e}_{t,\tau}$ is decreasing in τ . It is important to note, however, that e_τ is increasing over time, because ν_τ declines at a faster rate than the price of services increases. The hypothetical consumption

expenditure share is

$$\tilde{s}_{e,t,\tau} = \frac{\tilde{e}_{t,\tau}}{m_\tau}.$$

It is decreasing with τ since $\tilde{e}_{t,\tau}$ is decreasing and m_τ increasing.

Applying Roy's identity to $V(\tilde{e}_{t,\tau}, P_{g,\tau}, P_{s,\tau})$, the hypothetical share of goods on total consumption expenditure is

$$\tilde{s}_{g,t,\tau} := \frac{P_{g,\tau} \tilde{c}_{g,t,\tau}}{\tilde{e}_{t,\tau}} = \eta \left(\frac{\tilde{e}_{t,\tau}}{P_{s,\tau}} \right)^{-\chi} \left(\frac{P_{g,\tau}}{P_{s,\tau}} \right)^\gamma.$$

The hypothetical share of goods in total consumption may be growing or declining. Since $P_{g,\tau}/P_{s,\tau}$ is declining in τ and $\gamma > 0$, the relative price effect unambiguously reduces $\tilde{s}_{g,t,\tau}$. The income effect works in the opposite direction: $\tilde{e}_{t,\tau}/P_{s,\tau}$ is declining in τ and, with $\chi > 0$, this raises $(\tilde{e}_{t,\tau}/P_{s,\tau})^{-\chi}$, increasing the share of goods. This is the reverse of the standard case where both preferences and prices vary over time, in which rising income drives expenditure toward services.

Second stage. At the second stage, $\hat{m}_{t,z,\tau}$ is determined by equation (21). By construction, when using time- t preferences, as represented by ν_t , there exists a vector $(\hat{c}_{g,t,z,\tau}, \hat{c}_{s,t,z,\tau}, \hat{x}_{t,z,\tau})$ that solves the maximization problem:

$$\max_{\hat{c}_g, \hat{c}_s, \hat{x}} U(\hat{c}_{g,t,z,\tau}, \hat{c}_{s,t,z,\tau}) + \nu_t \hat{x}_{t,z,\tau}$$

subject to

$$\hat{m}_{t,z,\tau} = P_{g,z} \hat{c}_{g,t,z,\tau} + P_{s,z} \hat{c}_{s,t,z,\tau} + \hat{x}_{t,z,\tau}.$$

Using the indirect utility function (6), this problem reduces to:

$$\max_{\hat{e}, \hat{x}} V(\hat{e}_{t,z,\tau}, P_{g,z}, P_{s,z}) + \nu_t \hat{x}_{t,z,\tau}$$

subject to

$$\hat{m}_{t,z,\tau} = \hat{e}_{t,z,\tau} + \hat{x}_{t,z,\tau},$$

where $\hat{e}_{t,z,\tau} = P_{g,z} \hat{c}_{g,t,z,\tau} + P_{s,z} \hat{c}_{s,t,z,\tau}$.

The first-order condition with respect to $\hat{e}_{t,z,\tau}$ yields, after some algebra,

$$\hat{e}_{t,z,\tau} = (\nu_t P_{s,z}^\chi)^{\frac{1}{\chi-1}} := \hat{e}_{t,z}.$$

Notice that $\tilde{e}_{t,\tau}$ and $\hat{e}_{t,z}$ are formally identical.

Since the Bellman representation is quasilinear in $\hat{x}$, optimal consumption expenditure $\hat{e}$ does not depend on $\hat{m}$ and thus becomes independent of time τ . In other words, the household optimally chooses a constant hypothetical consumption expenditure $\hat{e}_{t,z}$, for all τ throughout the evaluated period, which depends only on the base preference parameter ν_t and base prices $P_{s,z}$.

The equivalent variation measure of the consumption expenditure share is

$$\hat{s}_{e,t,z,\tau} = \frac{\hat{e}_{t,z}}{\hat{m}_{t,z,\tau}}.$$

As long as hypothetical income grows with τ , the hypothetical consumption expenditure share is decreasing.

Applying Roy's identity to $V(\hat{e}_{t,z}, P_{g,z}, P_{s,z})$, which is independent of τ , gives the equivalent variation measure of the goods consumption share:

$$\hat{s}_{g,t,z} := \frac{P_{g,z} \hat{c}_{g,t,z}}{\hat{e}_{t,z}} = \eta \left(\frac{\hat{e}_{t,z}}{P_{s,z}} \right)^{-\chi} \left(\frac{P_{g,z}}{P_{s,z}} \right)^{\gamma},$$

the same for all τ throughout the evaluated period. Thus, the equivalent variation measure of the share of goods in total consumption is constant for all τ . Note also that $\hat{s}_{g,t,z}$ and $\tilde{s}_{g,t,\tau}$ are formally identical.

The constancy of $\hat{s}_{g,t,z}$ follows from two mutually reinforcing forces. First, the income effect is muted: quasilinearity pins down optimal consumption expenditure $\hat{e}_{t,z}$ independently of τ , so there is no income variation to shift the goods share. Second, the relative price effect is also muted: since prices are fixed at z , the term $(P_{g,z}/P_{s,z})^{\gamma}$ is constant throughout the evaluated period. With both channels shut down, $\hat{s}_{g,t,z}$ is invariant to τ .

Current-base Fisher–Shell index. For a fixed-base Fisher–Shell index, with base t , for all $\tau \in (t_0, t_1)$,

$$\hat{s}_{e,t,t,\tau} = \frac{e_t}{\hat{m}_{t,t,\tau}} \quad \text{and} \quad \hat{s}_{g,t,t} = \eta \left(\frac{e_t}{P_{s,t}} \right)^{-\chi} \left(\frac{P_{g,t}}{P_{s,t}} \right)^{\gamma} = s_{g,t}.$$

Then, for the current-base FS index the equivalent variation measure of the share of goods consumption is constant, equal to the share in current period s_{g,t_1} . The consumption expenditure share is declining, since e_{t_1} is constant but $\hat{m}_{t_1,t_1,\tau}$ is increasing with τ , converging from above to s_{e,t_1} as time converges to t_1 .

E Transformation Distortions in Consumption

This appendix summarizes how the benchmark [Herrendorf et al. \(2021\)](#) (HRV) environment changes when there are sector-specific distortions in the transformation of production output into usable consumption goods. The key feature of this extension is that the distortion affects the prices paid by households for final consumption uses, but leaves producer decisions and the investment block unchanged. Hence, the production side, the allocation of capital and labor across sectors, the investment aggregator, and the aggregate technology representation remain the same as in the benchmark model.

Throughout this appendix, labor is normalized to $L_t = 1$. Since the transformation wedge alters only the prices faced by households, it does not interact with the labor input, and the normalization is therefore without loss of generality: restoring L_t reinstates the factor $L_t^{1-\theta}$ in the aggregate production function and the term g_h in the growth rate of capital, exactly as in Section 2 of the paper, and leaves every statement below unchanged.

E.1 Transformation Technology and Prices

Let $P_{j,t}$ denote the producer price of sector j , $j \in \{g, s\}$, and let $P_{j,t}^c$ denote the price paid by households for usable consumption units. Delivering one unit of usable consumption in sector j requires $e^{\varphi_j t}$ units of raw output. Therefore,

$$P_{j,t}^c = e^{\varphi_j t} P_{j,t}, \quad j \in \{g, s\}. \quad (\text{OA.3})$$

When $\varphi_j = 0$, the benchmark model is recovered. The investment sector purchases raw output at producer prices $P_{j,t}$ and is therefore unaffected by the transformation wedge.

Because firms face producer prices, the sectoral value-added producer's problem is identical to the benchmark model. With common factor prices and a common capital share, the capital-labor ratio is equalized across sectors:

$$\frac{K_{g,t}}{L_{g,t}} = \frac{K_{s,t}}{L_{s,t}} = \frac{K_t}{L_t}.$$

It follows that relative producer prices remain determined by relative sectoral productivities:

$$\frac{P_{g,t}}{P_{s,t}} = \frac{A_{s,t}}{A_{g,t}}. \quad (\text{OA.4})$$

By contrast, relative expenditure prices satisfy

$$\frac{P_{g,t}^c}{P_{s,t}^c} = \frac{A_{s,t}}{A_{g,t}} e^{(\varphi_g - \varphi_s)t}. \quad (\text{OA.5})$$

Thus, the transformation wedge affects structural change only through the prices relevant for household consumption expenditure.

E.2 Investment Block and Aggregate Technology

The investment producer is unchanged. Normalizing the price of investment to one, the unit-cost condition is

$$1 = \frac{[\omega P_{g,t}^{1-\varepsilon} + (1-\omega)P_{s,t}^{1-\varepsilon}]^{1/(1-\varepsilon)}}{A_{x,t}}. \quad (\text{OA.6})$$

Combining this condition with (OA.4), sectoral producer prices are

$$P_{j,t} = \frac{\mathcal{A}_t}{A_{j,t}}, \quad j \in \{g, s\},$$

where

$$\mathcal{A}_t = A_{x,t} [\omega A_{g,t}^{\varepsilon-1} + (1-\omega)A_{s,t}^{\varepsilon-1}]^{1/(\varepsilon-1)}, \quad (\text{OA.7})$$

which is the investment-sector productivity index of equation (3) in the paper. The expenditure prices faced by households are therefore

$$P_{j,t}^c = e^{\varphi_j t} \frac{\mathcal{A}_t}{A_{j,t}}.$$

Since $P_{j,t}A_{j,t} = \mathcal{A}_t$ and the capital-labor ratio is common across sectors, the benchmark aggregation result continues to hold:

$$Y_t = \mathcal{A}_t K_t^\theta. \quad (\text{OA.8})$$

Factor prices are also unchanged relative to the benchmark model:

$$R_t = \theta \mathcal{A}_t K_t^{\theta-1}, \quad W_t = (1-\theta) \mathcal{A}_t K_t^\theta.$$

Thus, the transformation wedge does not affect production decisions, factor prices, or the aggregate production representation.

E.3 Resource Constraint and Aggregate Balanced Growth Path

The wedge changes the resource constraint for final consumption uses. Sectoral resource feasibility is

$$Y_{j,t} = X_{j,t} + e^{\varphi_{jt}} C_{j,t}.$$

Multiplying by producer prices and summing across sectors gives

$$Y_t = I_t + E_t^c, \quad E_t^c = P_{g,t}^c C_{g,t} + P_{s,t}^c C_{s,t}. \quad (\text{OA.9})$$

Aggregate output measured at producer prices is therefore allocated between investment and household consumption expenditure measured at expenditure prices.

The aggregate balanced growth path is governed by the productivity index $\mathcal{A}_t$. Denote by g_z the growth rate of any variable z . Along an ABGP, the rental rate is constant only if

$$g_K = \frac{g_A}{1 - \theta}.$$

Using (OA.8), output growth is

$$g_Y = g_A + \theta g_K.$$

Hence,

$$g_Y = g_K = \frac{g_A}{1 - \theta}. \quad (\text{OA.10})$$

The capital accumulation equation implies that I_t/K_t is constant along the ABGP, so $g_I = g_K$. Combining this result with (OA.9) implies

$$g_Y = g_I = g_K = g_{E^c}. \quad (\text{OA.11})$$

Thus, the transformation wedge changes the level and composition of consumption expenditure but not the growth rates of the aggregate objects along the ABGP.

Given an initial capital stock K_0 , the initial level of consumption expenditure that places the economy on the ABGP is

$$E_0^c = K_0 [\mathcal{A}_0 K_0^{\theta-1} - g_K - \delta]. \quad (\text{OA.12})$$

This condition is the same resource condition used in the benchmark model, except that consumption expenditure is evaluated at expenditure prices rather than producer prices. With labor growth restored, g_K is replaced by $g_K + n$, reproducing equation (12) of the paper in per capita terms.

E.4 Measurement

The empirical implementation requires distinguishing producer prices from expenditure prices. Producer prices are the prices relevant for firms' decisions and for the measurement of sectoral value added. Expenditure prices are the prices faced by households when purchasing usable consumption units. Sectoral productivities are therefore recovered from production data using sectoral real value added and factor inputs,

$$A_{j,t} = \frac{Y_{j,t}}{K_{j,t}^\theta L_{j,t}^{1-\theta}},$$

while aggregate productivity is recovered from the aggregate Solow residual,

$$\mathcal{A}_t = \frac{Y_t}{K_t^\theta}.$$

Observed relative prices correspond to expenditure prices. We normalize the service wedge to zero, $\varphi_s = 0$, and define

$$\zeta := \varphi_s - \varphi_g = -\varphi_g.$$

Using (OA.5), the ratio between the observed relative expenditure price and the model-implied relative producer price satisfies

$$\frac{P_{g,t}^c/P_{s,t}^c}{A_{s,t}/A_{g,t}} = e^{-\zeta t}. \quad (\text{OA.13})$$

We estimate ζ by fitting an exponential trend to this series. This procedure isolates the component of relative price movements that cannot be accounted for by sectoral productivity differences and attributes it to the transformation wedge. The estimate is

$$\zeta = 0.0137.$$

References

- Baqae, D. R. and Burstein, A. (2022). Welfare and output with income effects and taste shocks. *The Quarterly Journal of Economics*, 138(2):769–834.
- Boppart, T. (2014). Structural change and the Kaldor facts in a growth model with relative price effects and non-Gorman preferences. *Econometrica*, 82(6):2167–2196.
- Durán, J. and Licandro, O. (2025). Is the output growth rate in NIPA a welfare measure? *The Economic Journal*, 135(665):119–143.
- Herrendorf, B., Rogerson, R., and Valentinyi, Á. (2021). Structural change in investment and consumption—a unified analysis. *The Review of Economic Studies*, 88(3):1311–1346.